AAA WNYC Field Service Lightning

ERS Data Model Reference

Emergency Roadside Service - Salesforce Object Model

FSLAPP Backend Data Architecture

Version 1.0 - March 13, 2026

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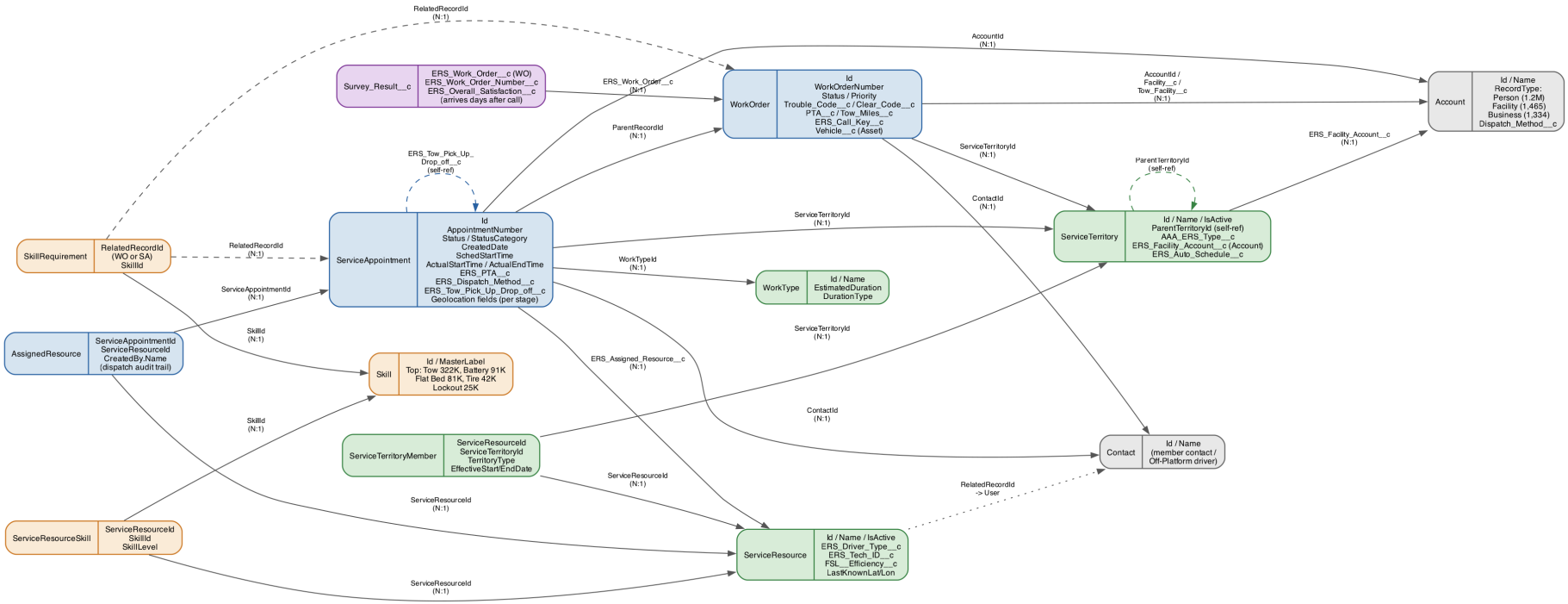
Color Legend

- Core ERS (ServiceAppointment, WorkOrder, AssignedResource)
- Scheduling (ServiceTerritory, ServiceResource, WorkType, STM)
- Skills (Skill, ServiceResourceSkill, SkillRequirement)
- Surveys (Survey_Result__c)
- CRM (Account, Contact)

Entity-Relationship Diagram



ServiceAppointment

Core ERS call record - 191 fields. Each emergency roadside service call is tracked as a ServiceAppointment linked to a WorkOrder, ServiceTerritory (garage), WorkType, and ServiceResource (driver).

Total fields:

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Salesforce primary key
AppointmentNumber	Auto Number	Appointment Number	N	Human-readable SA number
Status	Picklist	Status	N	None -> Scheduled -> Dispatched -> In Progress -> Completed / Canceled
StatusCategory	Picklist	Status Category	N	Salesforce standard category grouping
CreatedDate	DateTime	Created Date	N	When the SA record was created
SchedStartTime	DateTime	Scheduled Start	N	Scheduled start from optimizer
ActualStartTime	DateTime	Actual Start	N	CAUTION: Towbook writes future estimate, not real arrival
ActualEndTime	DateTime	Actual End	N	Completion timestamp
ServiceTerritoryId	Lookup	Service Territory	N	FK -> ServiceTerritory (garage)
WorkTypeId	Lookup	Work Type	N	FK -> WorkType (Tow, Battery, Tire, etc.)
ParentRecordId	Lookup	Parent Record	N	FK -> WorkOrder or WorkOrderLineItem
AccountId	Lookup	Account	N	FK -> Account (member person account)
ContactId	Lookup	Contact	N	FK -> Contact (member)
ERS_PTA__c	Number	Promised Time of Arrival	Y	PTA in minutes - key SLA metric
ERS_Dispatch_Method__c	Formula	Dispatch Method	Y	'Field Services' or 'Towbook'
ERS_Auto_Assign__c	Checkbox	Auto Assign	Y	UNRELIABLE - do NOT use for dispatch analysis
ERS_Cancellation_Reason__c	Picklist	Cancellation Reason	Y	Why SA was canceled
ERS_Facility_Decline_Reason__c	Picklist	Facility Decline Reason	Y	Why facility declined the call
Off_Platform_Driver__c	Lookup	Off-Platform Driver	Y	FK -> Contact (Towbook driver)
Off_Platform_Truck_Id__c	Text	Off-Platform Truck ID	Y	Towbook truck identifier string
ERS_Tow_Pick_Up_Drop_off__c	Lookup	Tow Pick-Up/Drop-Off	Y	Self-ref FK -> ServiceAppointment (tow pair)
ERS_Assigned_Resource__c	Lookup	Assigned Resource	Y	FK -> ServiceResource
Latitude (per stage)	Number	Geo Lat	Y	Lat at Accepted/Dispatched/EnRoute/OnLocation/Completed
Longitude (per stage)	Number	Geo Lon	Y	Lon at Accepted/Dispatched/EnRoute/OnLocation/Completed
FSL__Scheduling_Policy_Used__c	Lookup	Scheduling Policy	Y	FK -> FSL__Scheduling_Policy__c

Relationships

Field	Target Object	Card.	Description
ServiceTerritoryId	ServiceTerritory	N:1	Garage / territory
WorkTypeId	WorkType	N:1	Service type
ParentRecordId	WorkOrder / WOLI	N:1	Parent work order
AccountId	Account	N:1	Member account
ContactId	Contact	N:1	Member contact
ERS_Assigned_Resource__c	ServiceResource	N:1	Assigned driver
Off_Platform_Driver__c	Contact	N:1	Towbook driver
ERS_Tow_Pick_Up_Drop_off__c	ServiceAppointment	N:1	Self-ref tow pair

WorkOrder

Parent record of ServiceAppointment - 248 fields. Holds call-level details including trouble codes, vehicle, facility, and tow destination.

Total fields:

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
WorkOrderNumber	Auto Number	Work Order Number	N	Human-readable WO number
Status	Picklist	Status	N	Mirrors SA status lifecycle
Priority	Picklist	Priority	N	Call priority
ServiceTerritoryId	Lookup	Service Territory	N	FK -> ServiceTerritory
AccountId	Lookup	Account	N	FK -> Account (member)
ContactId	Lookup	Contact	N	FK -> Contact
WorkTypeId	Lookup	Work Type	N	Usually NULL - lives on WOLI/SA instead
ERS_Facility_Dispatch_Method__c	Formula	Facility Dispatch Method	Y	NOT groupable in SOQL (formula)
Facility__c	Lookup	Facility Account	Y	FK -> Account (facility)
Vehicle__c	Lookup	Vehicle	Y	FK -> Asset (member vehicle)
Tow_Facility__c	Lookup	Tow Destination	Y	FK -> Account (tow destination)
Trouble_Code__c	Text	Trouble Code	Y	ERS trouble code
Clear_Code__c	Text	Clear Code	Y	How the call was resolved
Resolution_Code__c	Text	Resolution Code	Y	Resolution detail
PTA__c	Number	PTA	Y	Promised Time of Arrival
Tow_Miles__c	Number	Tow Miles	Y	Distance for tow calls
Membership_ID__c	Text	Membership ID	Y	AAA member ID
ERS_Call_Key__c	Text	Call Key	Y	Unique ERS call key
ERS_Source_Call_Key__c	Text	Source Call Key	Y	Source system call key

Relationships

Field	Target Object	Card.	Description
ServiceTerritoryId	ServiceTerritory	N:1	Garage / territory
AccountId	Account	N:1	Member account
ContactId	Contact	N:1	Member contact
Facility__c	Account	N:1	Facility account
Tow_Facility__c	Account	N:1	Tow destination
Vehicle__c	Asset	N:1	Member vehicle

ServiceTerritory

Represents a garage or territory - 49 fields. Hierarchical via ParentTerritoryId. Linked to an Account for the facility.

Total fields.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
Name	Text	Territory Name	N	Garage / territory name
IsActive	Checkbox	Active	N	Whether territory is active
ParentTerritoryId	Lookup	Parent Territory	N	Self-ref FK -> ServiceTerritory
OperatingHoursId	Lookup	Operating Hours	N	FK -> OperatingHours
City	Text	City	N	Address city
State	Text	State	N	Address state
Latitude	Number	Latitude	N	Geo coordinate
Longitude	Number	Longitude	N	Geo coordinate
AAA_ERS_Type__c	Picklist	ERS Type	Y	Type of ERS territory
ERS_Facility_Account__c	Lookup	Facility Account	Y	FK -> Account (facility)
ERS_Auto_Schedule__c	Checkbox	Auto Schedule	Y	Whether auto-scheduling is enabled

Relationships

Field	Target Object	Card.	Description
ParentTerritoryId	ServiceTerritory	N:1	Self-ref hierarchy
OperatingHoursId	OperatingHours	N:1	Business hours
ERS_Facility_Account__c	Account	N:1	Facility account

ServiceResource

Driver / technician record. Tracks driver capabilities, GPS, and links to the Salesforce User record.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
Name	Text	Resource Name	N	Driver name
IsActive	Checkbox	Active	N	Whether resource is active
ERS_Driver_Type__c	Picklist	Driver Type	Y	Type of driver
ERS_Tech_ID__c	Text	Tech ID	Y	Technician identifier
FSL__Efficiency__c	Number	Efficiency	Y	FSL efficiency rating
LastKnownLatitude	Number	Last Known Lat	N	GPS latitude
LastKnownLongitude	Number	Last Known Lon	N	GPS longitude
RelatedRecordId	Lookup	Related Record	N	FK -> User

Relationships

Field	Target Object	Card.	Description
RelatedRecordId	User	N:1	Salesforce user record

ServiceTerritoryMember

Junction object linking ServiceResource (driver) to ServiceTerritory (garage). TerritoryType determines Primary/Secondary/Relocation assignment.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
ServiceResourceId	Lookup	Service Resource	N	FK -> ServiceResource
ServiceTerritoryId	Lookup	Service Territory	N	FK -> ServiceTerritory
TerritoryType	Picklist	Territory Type	N	Primary / Secondary / Relocation
EffectiveStartDate	DateTime	Effective Start	N	Membership start
EffectiveEndDate	DateTime	Effective End	N	Membership end

Relationships

Field	Target Object	Card.	Description
ServiceResourceId	ServiceResource	N:1	Driver
ServiceTerritoryId	ServiceTerritory	N:1	Garage / territory

AssignedResource

Junction linking a ServiceResource to a ServiceAppointment. CreatedBy.Name is the REAL audit trail for who dispatched the call - more reliable than ERS_Auto_Assign__c.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
ServiceAppointmentId	Lookup	Service Appointment	N	FK -> ServiceAppointment
ServiceResourceId	Lookup	Service Resource	N	FK -> ServiceResource
CreatedBy.Name	Text	Created By	N	WHO dispatched - the real audit trail

Relationships

Field	Target Object	Card.	Description
ServiceAppointmentId	ServiceAppointment	N:1	The call
ServiceResourceId	ServiceResource	N:1	The driver

WorkType

Defines service types: Tow, Battery, Tire Change, Lockout, Flat Bed, etc. Referenced by ServiceAppointment and WorkOrderLineItem.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
Name	Text	Work Type Name	N	e.g. Tow, Battery, Tire Change, Lockout, Flat Bed
EstimatedDuration	Number	Estimated Duration	N	Default duration for this service
DurationType	Picklist	Duration Type	N	Minutes / Hours

Skill

Represents a capability (55 skills total). Top skills by volume: Tow (322K), Battery (91K), Flat Bed (81K), Tire (42K), Lockout (25K).

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
MasterLabel	Text	Skill Name	N	e.g. Tow, Battery, Flat Bed, Tire, Lockout

ServiceResourceSkill

Junction linking a Skill to a ServiceResource (driver). Defines what services a driver can perform.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
ServiceResourceId	Lookup	Service Resource	N	FK -> ServiceResource
SkillId	Lookup	Skill	N	FK -> Skill
SkillLevel	Number	Skill Level	N	Proficiency level

Relationships

Field	Target Object	Card.	Description
ServiceResourceId	ServiceResource	N:1	Driver
SkillId	Skill	N:1	Capability

SkillRequirement

Defines what skills are needed for a WorkOrder or ServiceAppointment. Used by the optimizer to match drivers to calls.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
RelatedRecordId	Lookup	Related Record	N	FK -> WorkOrder or ServiceAppointment
SkillId	Lookup	Skill	N	FK -> Skill

Relationships

Field	Target Object	Card.	Description
RelatedRecordId	WorkOrder / SA	N:1	The call requiring the skill
SkillId	Skill	N:1	Required capability

Survey_Result__c

Custom object for CSAT survey results. Linked to WorkOrder. Surveys arrive days after the call completes - not real-time.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
ERS_Work_Order__c	Lookup	Work Order	Y	FK -> WorkOrder
ERS_Work_Order_Number__c	Text	Work Order Number	Y	Text match field for WO number
ERS_Overall_Satisfaction__c	Picklist	Overall Satisfaction	Y	Totally Satisfied / Satisfied / etc.

Relationships

Field	Target Object	Card.	Description
ERS_Work_Order__c	WorkOrder	N:1	The call being rated

Account

Person Accounts (1.2M members), Facility accounts (1,465 garages), Business accounts (1,334). RecordType determines the type.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
Name	Text	Account Name	N	Member or facility name
RecordTypeId	Lookup	Record Type	N	Person Account / Facility / Business
Dispatch_Method__c	Picklist	Dispatch Method	Y	On Facility accounts - dispatch routing

Contact

Contact records for members and off-platform (Towbook) drivers.

Field Name	Type	Label	Cust?	Key Notes
Id	ID	Record ID	N	Primary key
Name	Text	Contact Name	N	Full name
AccountId	Lookup	Account	N	FK -> Account

Relationships

Field	Target Object	Card.	Description
AccountId	Account	N:1	Parent account

Data Quality & Gotchas

The following data quality issues have been discovered during development and analysis of the FSLAPP backend. These are critical for anyone writing SOQL queries, building reports, or extending the application.

1 Towbook ActualStartTime is NOT real arrival

The Towbook integration writes a future estimated arrival time into ActualStartTime - it is NOT the actual on-scene time. Verified via ServiceAppointmentHistory analysis (March 2026). For accurate arrival times on Towbook calls, use the history object's 'In Progress' status transition timestamp.

2 ERS_Auto_Assign__c is unreliable

This checkbox does not reliably indicate whether a call was auto-dispatched or manually dispatched. For accurate dispatch method analysis, use AssignedResource.CreatedBy.Name instead - 'FSL Automation' indicates auto-dispatch; any other name indicates manual dispatch.

3 SA.Description is not filterable

The Description field on ServiceAppointment cannot be used in a SOQL WHERE clause. You can SELECT it but cannot filter on it.

4 Formula fields are NOT groupable in SOQL

Fields like ERS_Dispatch_Method__c and ERS_Facility_Dispatch_Method__c are formula fields and cannot be used in GROUP BY clauses. You must group by the underlying source fields or use post-query aggregation.

5 WorkType lives on WOLI/SA, not WorkOrder

WorkOrder.WorkTypeId is usually NULL. The actual work type is set on the WorkOrderLineItem and/or ServiceAppointment. Always join via SA.WorkTypeId.

6 Tow Drop-Off SAs must be excluded from response time analysis

Tow calls create two SAs: a Pick-Up and a Drop-Off. The Drop-Off SA's times reflect the tow delivery, not a separate ERS response. Exclude Drop-Off SAs (linked via ERS_Tow_Pick_Up_Drop_off__c) when calculating PTA compliance, response times, or call volumes.

7 Survey data arrives days after the call

Survey_Result__c records are not created at call completion. They arrive asynchronously days later. Any real-time dashboard showing CSAT must account for this lag. Match surveys to calls via ERS_Work_Order__c or ERS_Work_Order_Number__c.